

TOSHIBA STORAGE DEVICE DIVISION

MK3019GAXB/MK4019GAXB HDD INSTALLATION NOTES

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GENERAL DESCRIPTION

Toshiba's MK3019GAXB and MK4019GAXB (HDD2175/2174) hard disk drives are a series of fast, high-capacity 2.5" Winchester disk drives which comply with ATA-5 standards and require only a simple adapter board for interfacing to standard IDE cables.

SPECIFICATIONS	MK3019GAXB	MK4019GAXB
	HDD2175	HDD2174
FORMATTED CAPACITY	30.0056GB	40.0078GB
HEIGHT	9.5mm	9.5mm
WEIGHT	3.563 ounces (101grams)	3.563 ounces (101grams)
NO. OF DISKS (PLATTERS)	2	2
NO. OF DATA HEADS	3	4

FUNCTIONAL SPECIFICATION

LOGICAL DRIVE PARAMETERS	MK3019GAXB	MK4019GAXB
	HDD2175	HDD2174
NO. OF CYLINDERS (user)	33,440	
BYTES PER SECTOR	512	
BUFFER SIZE	16,384Kbytes	
ROTATION SPEED ($\pm 0.1\%$)	5,400rpm	
TRANSFER RATE		
INTERNAL (Mbits/sec.)	200.8~333.2	
NO. OF CYLINDERS (logical)	16,383	
NO. OF HEADS (logical)	16	
NO. OF SECTORS/TRACK	63	
HOST TRANSFER RATE		
PIO/ULTRA DMA (ATA MODE)	16.6/100MB/s	
SEEK TIME		
AVERAGE	12ms	
TRACK-TO-TRACK	2ms	
MAXIMUM	22ms	
TEMPERATURE		
OPERATING	41°F - 104°F (5°C - 40°C)	
NON-OPERATING	-4°F - 140°F (-20°C - 60°C)	
HUMIDITY		
OPERATING/NON-OPERATING	8 - 90% R.H. (No condensation)	
SHIPPING	5 - 90% R.H. (No condensation)	
WET BULB	29°C(operating)/40°C(non-operating)	

UNPACKING PROCEDURE

Visually inspect the shipping container prior to unpacking for any signs of damage to the container or its contents (the carrier is responsible for any damage incurred during shipment). The drive is a precision device and even a small drop onto any surface can cause damage. Prior to opening the anti-static bag, it is recommended that the user ground himself with a ground strap or by touching the PC chassis or other metal object. Remove the drive from the anti-static bag and check it for damage. Place the drive on the anti-static bag while configuring jumpers to avoid any electrostatic discharge. Save the shipping container and packing material for possible use later.

DRIVE CONFIGURATION

Toshiba's 2.5" hard disk drives can be configured as either master or slave units. Master configuration is used for all single drive applications, and master or slave configuration (only one of each per port) is used for two drive applications. Use the information in the following table before setting M/S drive.

Master configuration is obtained by setting jumpers B, C & D open (no jumper). Slave configuration is obtained by setting jumpers C-D. When B-D jumper is installed, the drive is configured as cable select. If pin 28 = Low, the drive is master. If pin 28 = High, the drive is slave.

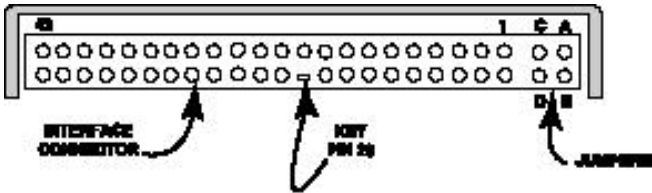


FIGURE 1 – 2.5" HDD JUMPER LOCATIONS

JUMPER	P28	DRIVE
No Jumper	-	Master Drive
C-D Jumper	-	Slave Drive
B-D Jumper	LOW	Master Drive
B-D Jumper	HIGH	Slave Drive
A-B Jumper	-	Slave Drive
A-C Jumper	-	Prohibit



FIGURE 2 – MOUNTING HOLE LOCATIONS

INSTALLATION INSTRUCTIONS

1. Determine mounting configuration (can be mounted in either a horizontal or vertical orientation).
2. Configure drive for system application.
3. Configure the adapter board for the specific system application (if required).
4. Install adapter board into an unused PC/AT 16 bit slot (if required).
5. Install the I/F cable to the System's 44 pin connector port or adapter board. Ensure Pin 1 is oriented correctly. (Pin 1 on the cable is usually identified by a red or blue stripe.)
6. Set correct drive type in system CMOS.
7. Refer to applicable manuals for software installation instructions.

TECHNICAL SUPPORT

Should you require technical support, contact your computer distributor. If your distributor is unable to answer your questions, have them call Toshiba Storage Device Division Technical Support, on your behalf.

TOSHIBA - STORAGE DEVICE DIVISION
TECHNICAL SUPPORT - 805.644.6350

You can also visit Toshiba's web site at: <http://www.sdd.toshiba.com>

CONNECTOR CABLING INFORMATION

DRIVE INTERFACE SIGNALS			
PIN	SIGNAL	PIN	SIGNAL
1	-RESET	2	GROUND
3	DD 7	4	DD 8
5	DD 6	6	DD 9
7	DD 5	8	DD 10
9	DD 4	10	DD 11
11	DD 3	12	DD 12
13	DD 2	14	DD 13
15	DD 1	16	DD 14
17	DD 0	18	DD 15
19	GROUND	20	KEY
21	DMARQ	22	GROUND
23	-DIOW	24	GROUND
	STOP		
25	-DIOR	26	GROUND
	-DMARDY		
	HSTROBE		
27	IORDY	28	CSEL
	-DMARDY		
	-DSTROBE		
29	-DMACK	30	GROUND
31	INTRQ	32	- IOCS16
33	DA1	34	- PDIAG
35	DA0	36	DA 2
37	- CS0	38	- CS1
39	- DASP	40	GROUND
41	+5V (LOGIC)	42	+5V (MOTOR)
43	GROUND	44	RESERVED
Note: Symbol (–) in front of signal indicates negative logic			